Introduction to Cryptography

Lecture 36: RSA Key Generation

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Learning Outcomes

- List the steps of RSA key generation.
- Calculate n and $\phi(n)$ from two primes.
- Choose a valid public exponent.
- Compute the private exponent using the Extended Euclidean Algorithm.

RSA Overview

RSA was developed by Rivest, Shamir and Adleman. It operates on integer messages $0 \leq M < n$.

$$\text{Public key} = (e, n), \quad \text{Private key} = (d, n)$$

Key Generation Algorithm

1. Select distinct primes p, q.
2. Compute $n = pq$.
3. Compute $\phi(n) = (p-1)(q-1)$.
4. Choose $1 < e < \phi(n)$ with $\gcd(e, \phi(n)) = 1$.
5. Compute $d = e^{-1} \bmod \phi(n)$.

Worked Example: Choose Primes

$$p = 17, \quad q = 11$$

$$n = pq = 17 \times 11 = \boxed{187}$$

$$\phi(n) = 16 \times 10 = \boxed{160}$$

Choose Public Exponent

Choose $e = 7$. Verify:

$$\gcd(7, 160) = 1$$

Hence $e = 7$ is valid.

$$PU = (e, n) = \boxed{(7, 187)}$$

Find Private Exponent

We need

$$7d \equiv 1 \pmod{160}.$$

Since

$$160 = 22(7) + 6, \quad 7 = 1(6) + 1,$$

$$1 = 7 - (160 - 22 \cdot 7) = 23 \cdot 7 - 160.$$

Thus $d = \boxed{23}$.

Verify the Key Pair

$$ed = 7 \times 23 = 161 = 160 + 1$$

$$ed \bmod \phi(n) = 1$$

$$\boxed{PU = (7, 187), \quad PR = (23, 187)}$$

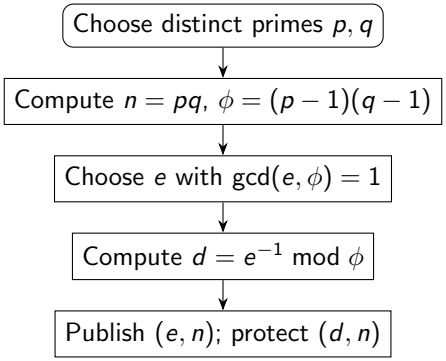
Why Must e Be Coprime?

The modular inverse $e^{-1} \pmod{\phi(n)}$ exists exactly when

$$\gcd(e, \phi(n)) = 1.$$

Example: $e = 10$ is invalid for $\phi(n) = 160$ because $\gcd(10, 160) = 10$.

RSA Key Generation Flow



Practical Key Generation

- Generate large random candidate primes.
- Use reliable primality testing such as Miller–Rabin.
- Use a cryptographically secure random generator.
- A common public exponent is 65537.
- Modern RSA uses appropriately sized moduli; classroom primes are only illustrative.

Security Considerations

- Never disclose p, q, d, or secret CRT parameters.
- Small textbook RSA moduli can be factored instantly.
- Do not use raw textbook RSA for real messages.
- Use standardized RSA-OAEP for encryption or RSA-PSS for signatures.

Student Exercise

Generate RSA keys for $p = 13$, $q = 17$, and $e = 5$. Find n, $\phi(n)$ and d.

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Generate RSA keys for $p = 13$, $q = 17$, and $e = 5$. Find n, $\phi(n)$ and d.

$$n = 221, \quad \phi(n) = 192, \quad d = 77$$

$$5 \times 77 = 385 \equiv 1 \pmod{192}.$$

Recap

$$\boxed{n = pq, \quad \phi(n) = (p-1)(q-1)}$$

$$\boxed{\gcd(e, \phi(n)) = 1, \quad ed \equiv 1 \pmod{\phi(n)}}$$

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